OS W7 Practical Assignment

Inode Simulation — Sector Cost Calculation & Addressing Method

1. Objective

Extend **inode_sim.sh** to calculate the byte count and sector cost for several file sizes, and decide whether the 12 direct pointers of an inode are enough to address each file.

2. Assumptions and Formulas

Parameter / Formula	Value
Sector size	512 bytes
Direct pointers in inode	12
Maximum direct-addressable size	12 x 512 = 6,144 bytes
Byte count	KB = 1024 B, MB = 1024 x 1024 B, GB = 1024 x 1024 x 1024 B
Sector count (rounded up)	ceil(bytes / 512) = (bytes + 511) / 512
Addressing method	bytes > 6,144 → INDIRECT ADDRESSING NEEDED; otherwise DIRECT ADDRESSING SUFFICIENT

3. Bash Script (inode_sim.sh)

```
#!/bin/bash

# ============================================================
# Assignment: Inode Simulation - Sector Cost Calculation
# Description: Calculates the sector cost for different
#              file sizes and determines the addressing method.
# ============================================================

# System parameters
SECTOR_SIZE=512
DIRECT_POINTERS=12
MAX_DIRECT_SIZE=$((SECTOR_SIZE * DIRECT_POINTERS))

# Function to calculate sectors required for a file
calculate_sectors() {
    local bytes=$1

    # Round up to account for a partially occupied sector
    echo $(((bytes + SECTOR_SIZE - 1) / SECTOR_SIZE))
}

# Function to determine the addressing method
get_addressing_method() {
    local bytes=$1

    if (( bytes <= MAX_DIRECT_SIZE )); then
        echo "DIRECT ADDRESSING SUFFICIENT"
    else
        echo "INDIRECT ADDRESSING NEEDED"
    fi
}

# File sizes in bytes (using binary units)
KB=$((1024))
```

```
MB=$((1024 * 1024))
GB=$((1024 * 1024 * 1024))

# Display assignment information
echo
echo "============================================================="
echo "          INODE SIMULATION - SECTOR COST CALCULATION"
echo "============================================================="
echo
echo "Sector Size          : $SECTOR_SIZE bytes"
echo "Direct Pointers      : $DIRECT_POINTERS"
echo "Maximum Direct Size  : $MAX_DIRECT_SIZE bytes"
echo

# Print the table header
printf "%-12s | %-15s | %-15s | %-30s\n" \
    "File Size" "Byte Count" "Sector Count" "Addressing Method"

printf '%s\n' \
    "-------------+-----------------+-----------------+--------------------------------"

# File sizes to evaluate
file_sizes=("1 KB" "100 KB" "1 MB" "1 GB" "10 GB")
byte_counts=(
    "$KB"
    "$((100 * KB))"
    "$MB"
    "$GB"
    "$((10 * GB))"
)

# Calculate and display results
for i in "${!file_sizes[@]}"; do
    bytes=${byte_counts[$i]}

    sectors=$(calculate_sectors "$bytes")
    method=$(get_addressing_method "$bytes")

    printf "%-12s | %-15s | %-15s | %-30s\n" \
        "${file_sizes[$i]}" "$bytes" "$sectors" "$method"
done

# Display the conclusion
echo
echo "============================================================="
echo "CONCLUSION"
echo "============================================================="
echo "Direct addressing can handle files up to $MAX_DIRECT_SIZE bytes."
echo "Files exceeding this limit require indirect addressing."
echo "============================================================="
echo
```

4. Terminal Execution and Output

```
$ chmod +x inode_sim.sh
$ ./inode_sim.sh
============================================================
          INODE SIMULATION - SECTOR COST CALCULATION
============================================================

Sector Size          : 512 bytes
Direct Pointers      : 12
Maximum Direct Size  : 6144 bytes

File Size    | Byte Count      | Sector Count    | Addressing Method
-------------+-----------------+-----------------+--------------------------------
1 KB         | 1024            | 2               | DIRECT ADDRESSING SUFFICIENT
100 KB       | 102400          | 200             | INDIRECT ADDRESSING NEEDED
1 MB         | 1048576         | 2048            | INDIRECT ADDRESSING NEEDED
1 GB         | 1073741824      | 2097152         | INDIRECT ADDRESSING NEEDED
10 GB        | 10737418240     | 20971520        | INDIRECT ADDRESSING NEEDED

============================================================
CONCLUSION
============================================================
Direct addressing can handle files up to 6144 bytes.
Files exceeding this limit require indirect addressing.
============================================================
```

5. Result Table

File Size	Byte Count	Sector Count	Addressing Method
1 KB	1,024	2	DIRECT ADDRESSING SUFFICIENT
100 KB	102,400	200	INDIRECT ADDRESSING NEEDED
1 MB	1,048,576	2,048	INDIRECT ADDRESSING NEEDED
1 GB	1,073,741,824	2,097,152	INDIRECT ADDRESSING NEEDED
10 GB	10,737,418,240	20,971,520	INDIRECT ADDRESSING NEEDED

6. Explanation

Sector count. A file always occupies whole sectors, so the byte count is divided by 512 and rounded up. For example, 1 KB = 1024 bytes needs 1024 / 512 = 2 sectors, and 100 KB = 102,400 bytes needs 200 sectors.

Addressing method. With 12 direct pointers of one sector each, an inode can reach at most 12 x 512 = 6,144 bytes directly. Only the 1 KB file (1,024 bytes) fits within this limit. The 100 KB, 1 MB, 1 GB and 10 GB files exceed 6,144 bytes, so they need indirect pointers (single, double or triple indirect blocks).

Implementation notes. The script uses two functions, *calculate_sectors* (ceiling division using integer arithmetic) and *get_addressing_method* (threshold comparison with MAX_DIRECT_SIZE), and prints the result in a formatted table with *printf*.

7. Conclusion

Direct addressing is sufficient only for files up to 6,144 bytes. Every larger file, including 100 KB, 1 MB, 1 GB and 10 GB, requires indirect addressing, and the sector cost grows linearly with file size.